

Auditory Contrast Network toward Text-Free Prominence Detection

Can local auditory contrast replace large speech representations?

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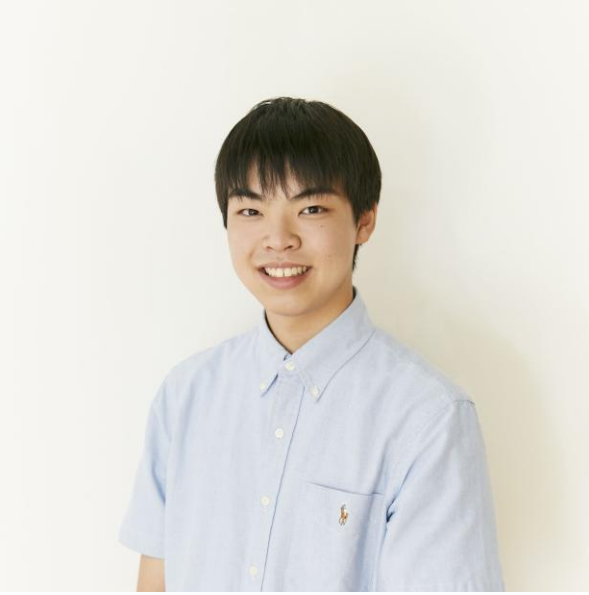
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Self-Introduction

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- Kosuke Shimizu
- B3. University of Tsukuba
- Human-Computer Interaction, VR Research
- First time to participate in Interspeech conference!

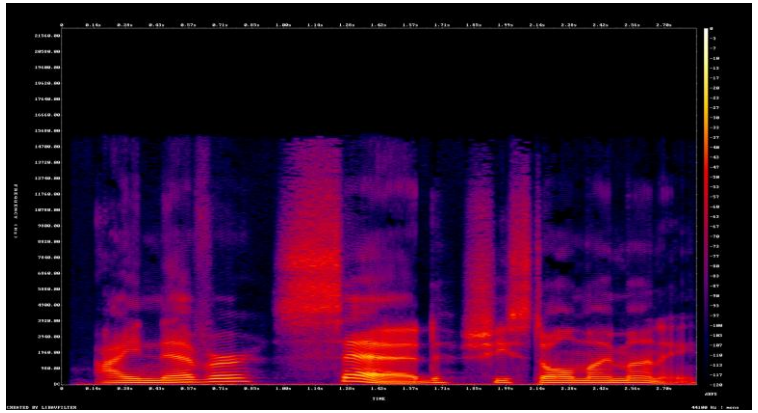


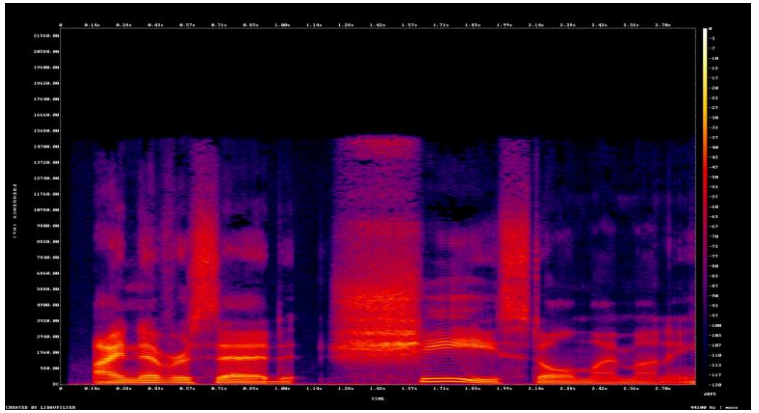


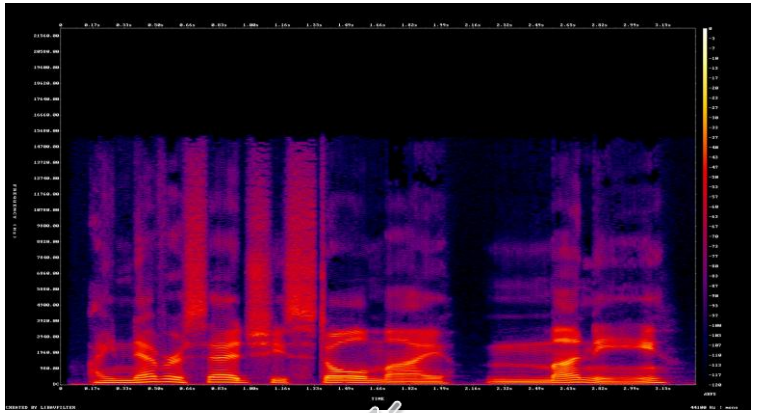
What is Prosodic Prominence?

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- Prominence is the perceived importance or emphasis of a word in speech.
  - I never SAID she stole my money. (I only implied it.)
  - I never said SHE stole my money. （Someone stole it.)
  - I never said she stole my MONEY. (She stole something.)



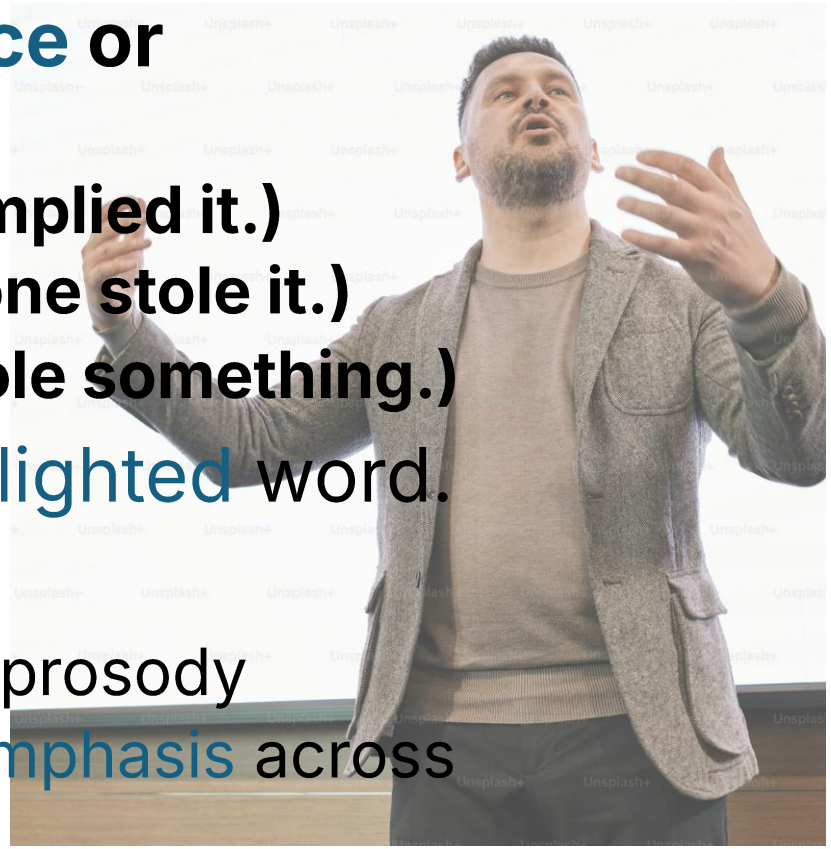




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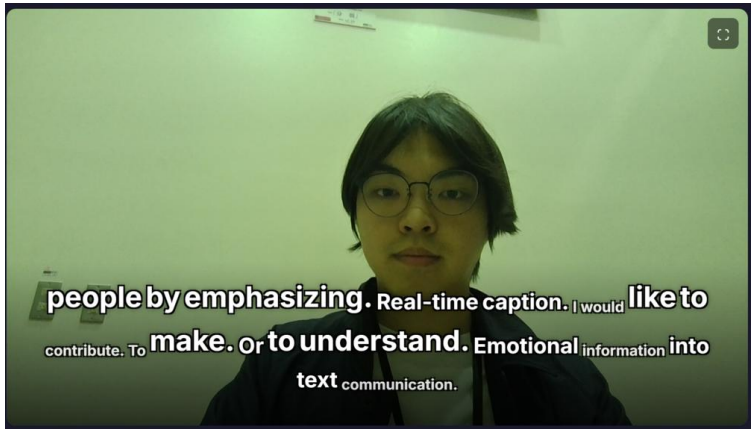
- **Prominence is the perceived importance or emphasis of a word in speech.**
  - **I never SAID she stole my money. (I only implied it.)**
  - **I never said SHE stole my money. (Someone stole it.)**
  - **I never said she stole my MONEY. (She stole something.)**
- Same sentence structure, different highlighted word.
- **Why So Matter**
  - **Text-to-speech:** controlling emphasis and prosody
  - **Speech-to-speech systems:** preserving emphasis across voices or languages



Using Prominence Information

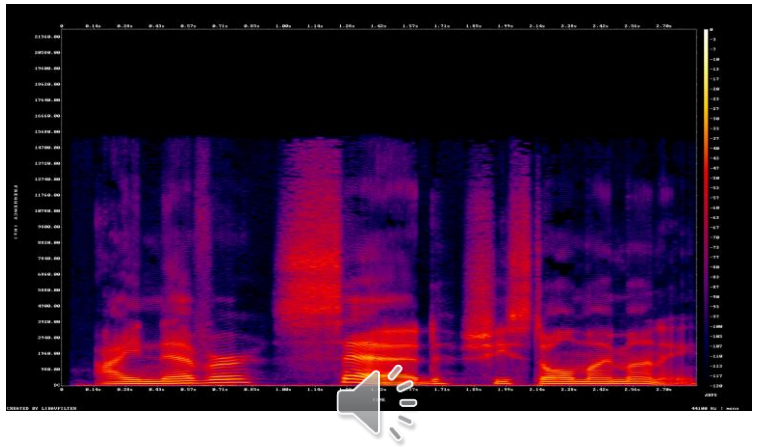
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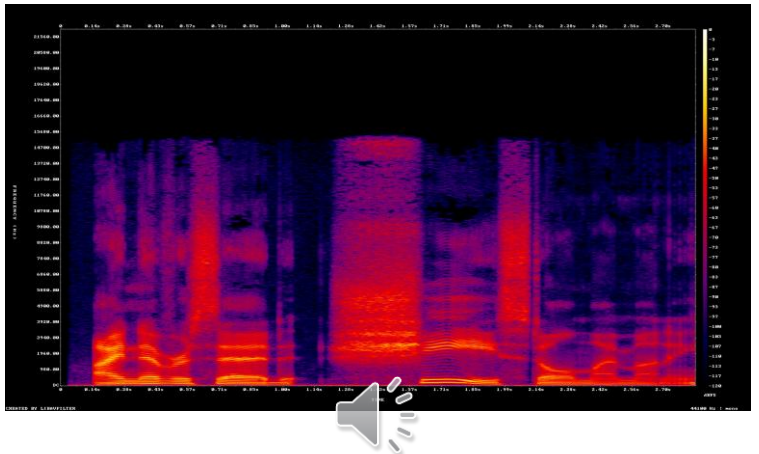
- **Our goal is to convey prosodic information as well as text information in real-time transcription for Deaf people.**
  - i.e.) Augmented Reality Glasses, Realtime Transcription
- System should be fast enough to support live communication.

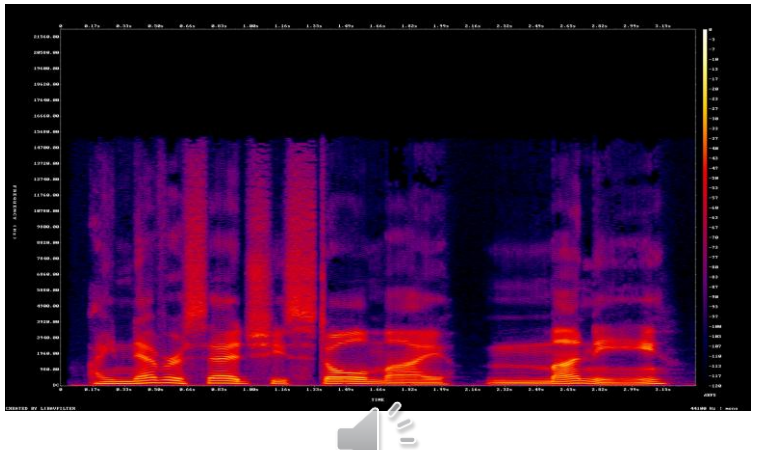


Prominence is Relational

- Prominence is not an absolute property of a word.
  - A 300-ms word may sound ordinary in slow speech, but prominent among 150-ms words.
- Contrast with adjacent terms such as duration, intensity, and spectral shape contributes to prominence.
- We perceive a word relative to its acoustic context.



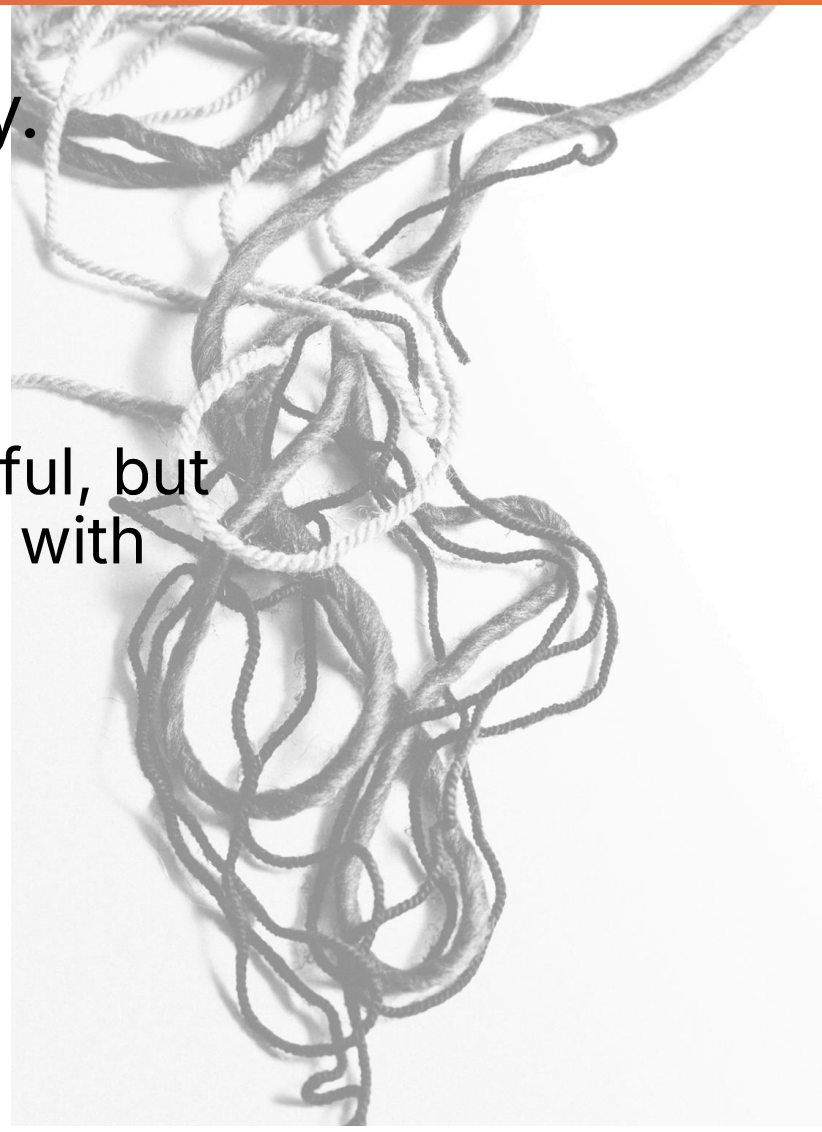




Relation Based Mathematical Modeling

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- Current models learn prominence implicitly.
  - Speech→wav2vec→Prominence
- **Can we explicitly model the auditory computation itself?**
  - CNN and wav2vec2, among others, are powerful, but they do not explicitly incorporate comparisons with adjacent words into their architecture.



Relation Based Mathematical Modeling

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We are inspired by three psychoacoustic characteristics into the model

- Pairwise contrast

Perception depends on **relative change**, not just absolute values(Cole et al. (2010)).

- Cue Specific-processing

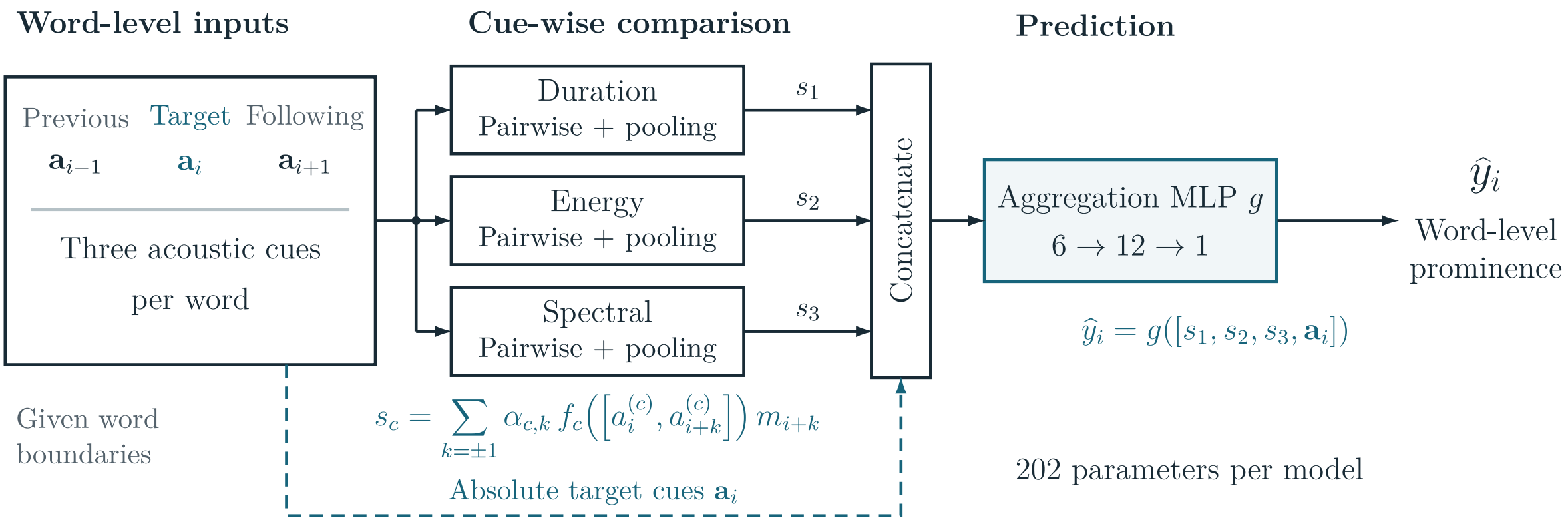
Duration, energy, and spectral cues are compared **separately**(Sluijter & van Heuven (1996)).

- Forward dominance

Recent auditory context may affect perception **asymmetrically**(Ulanovsky et al. (2003)).

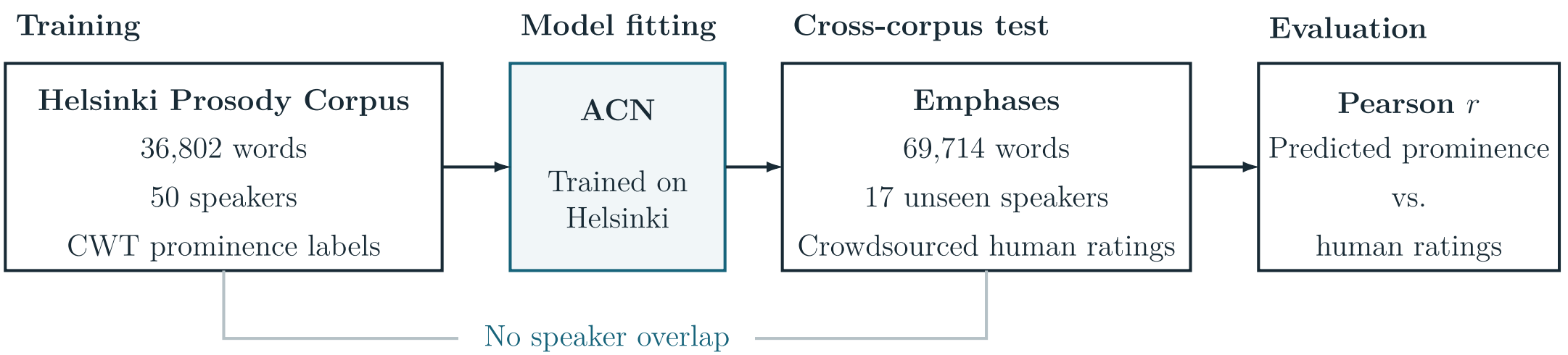
Auditory Contrast Network

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s_c: contrast score for cue c, $a_i^{(c)}$, $a_{i+k}^{(c)}$: target and neighboring cue values, f_c: cue−specific MLP $(2 \to 8 \to 1)$
$\alpha_{c,k}$: learned directional weight, $m_{i+k} \in \{0,1\}$: neighbor−presence mask, $k = -1$: preceding word, $k = +1$: following word

How to Evaluate



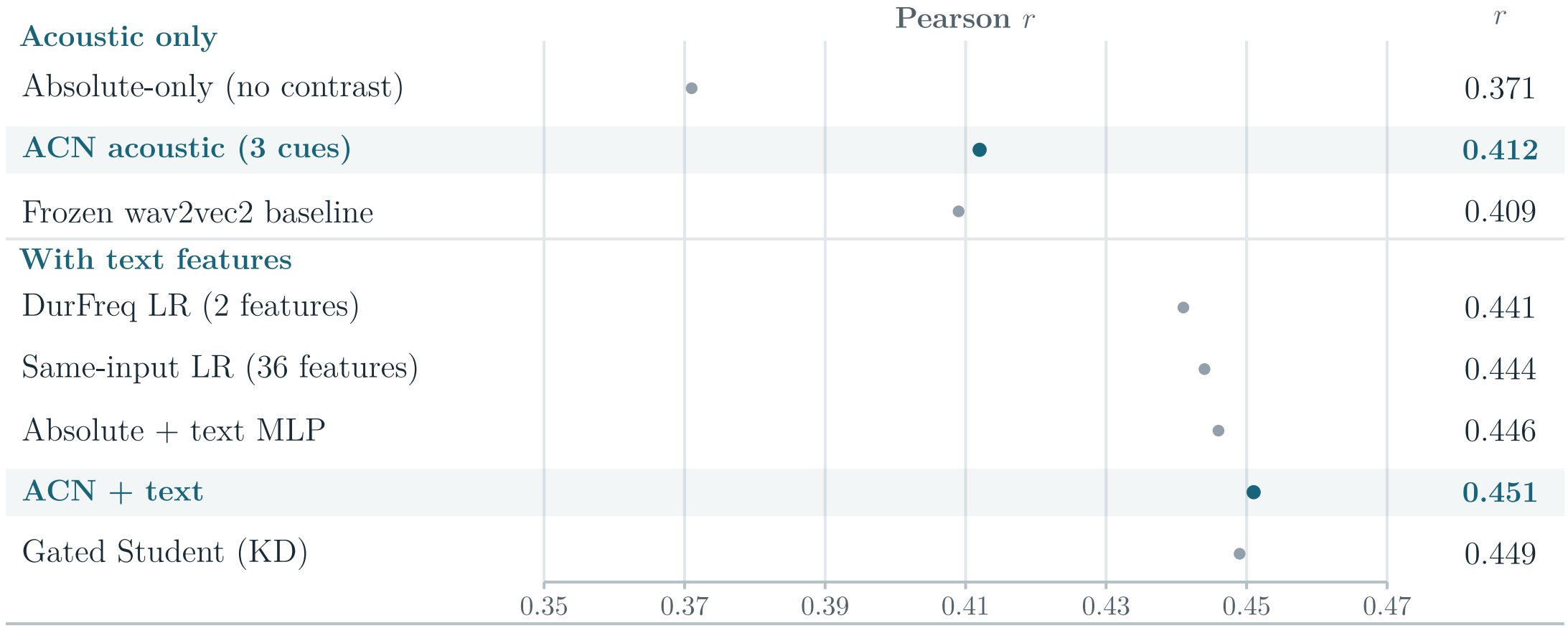
Comparison conditions

Absolute-only	wav2vec2-base	With text features	ACN + text
Same 3 acoustic cues	Frozen encoder	DurFreq LR, Same-input LR	Acoustic cues
No pairwise contrast	Layer-9 pooled embeddings	Gated Student (KD)	+ text features

ACN training: Adam, learning rate 0.003, batch size 4096, early stopping

Main Result

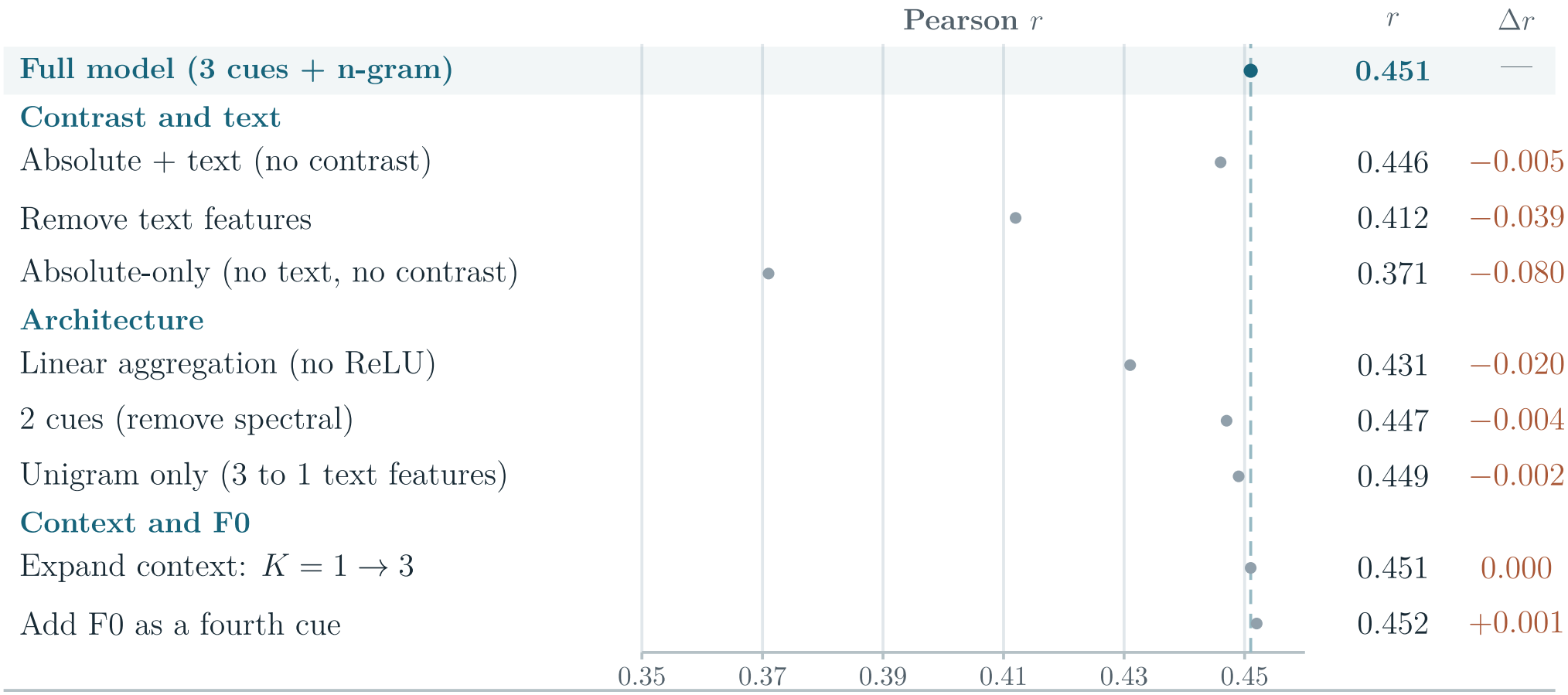
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ACN: 202 / 238 parameters per model (acoustic / + text).
ACN scores use a 3-model ensemble; wav2vec2 uses a frozen 94.6M-parameter encoder + trained head.
Reported CPU time: 1.6 vs. 194 ms/word (≈120×); single thread, excluding word segmentation.

Why does it work?

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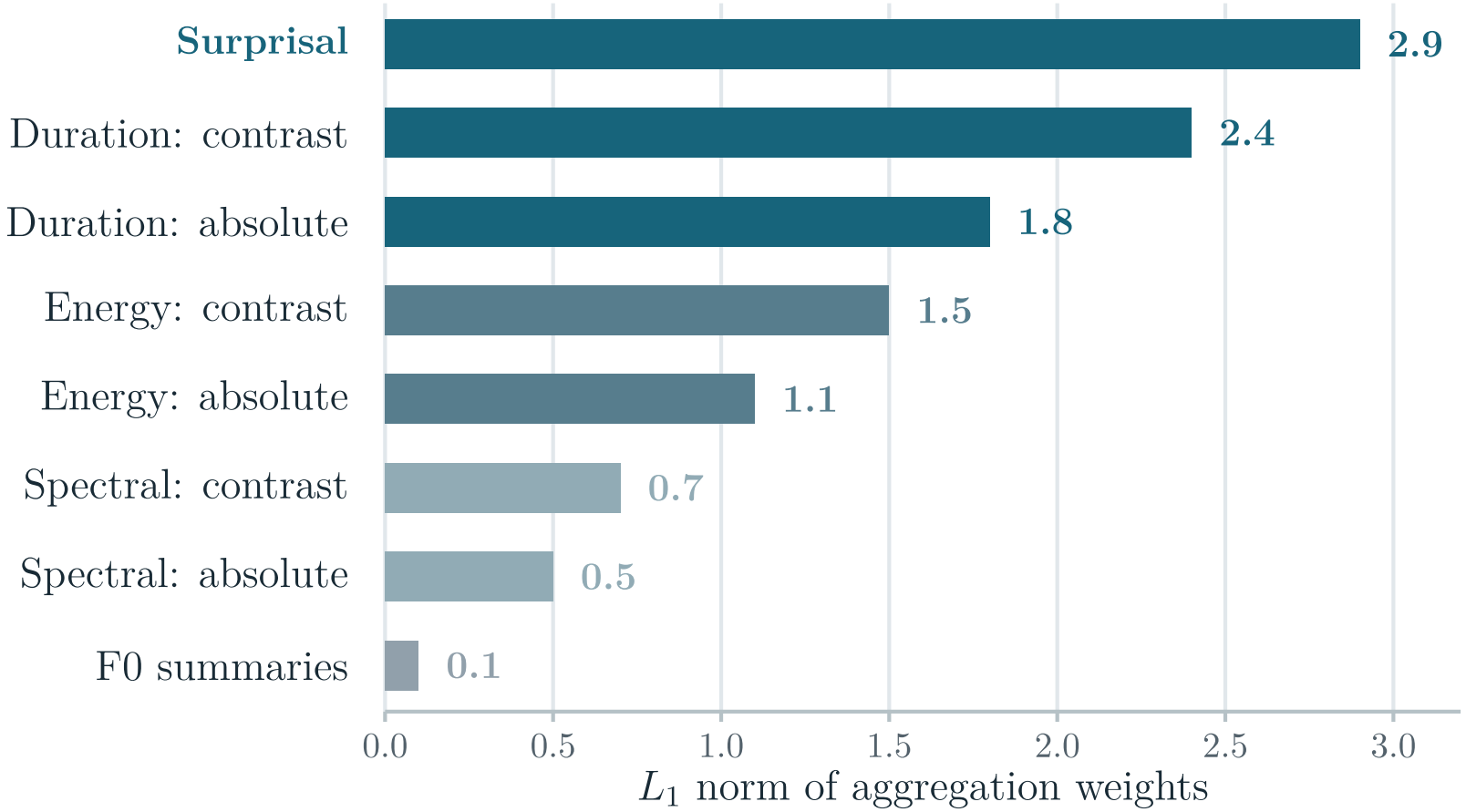


Pairwise contrast: $\Delta r = +0.041$ without text; $+0.005$ with text.

All scores: 3-model ensembles. Δr is relative to the full model; dashed line marks $r = 0.451$.
No gain from wider context; tested F0 summaries add 0.001 in this evaluation.

Discussion

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Largest overall

Surprisal

Acoustic cues

Duration
> Energy
> Spectral

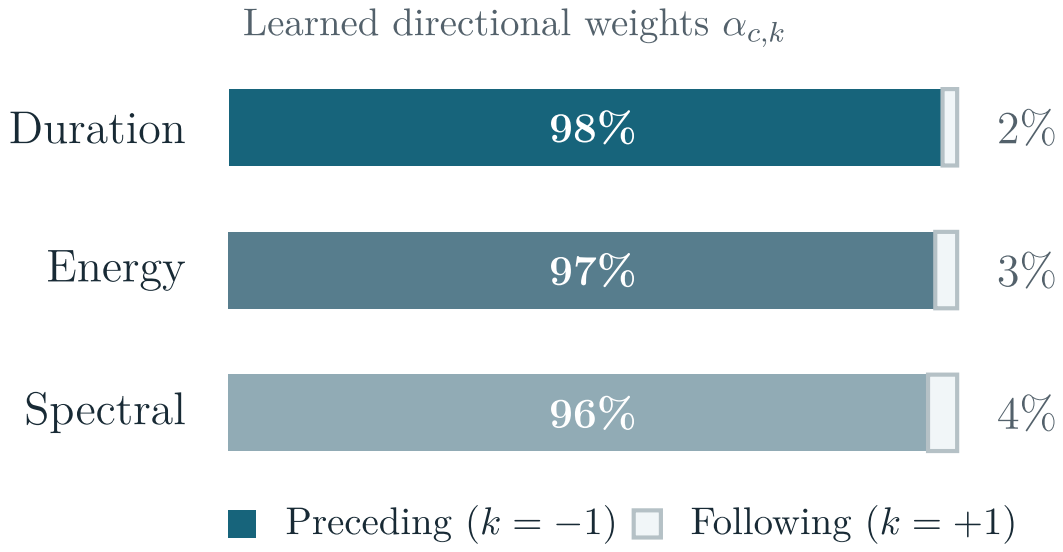
Small weights for the tested F0 summaries

Best single seed. Weight norms depend on input scale and are a proxy for model influence.

Discussion

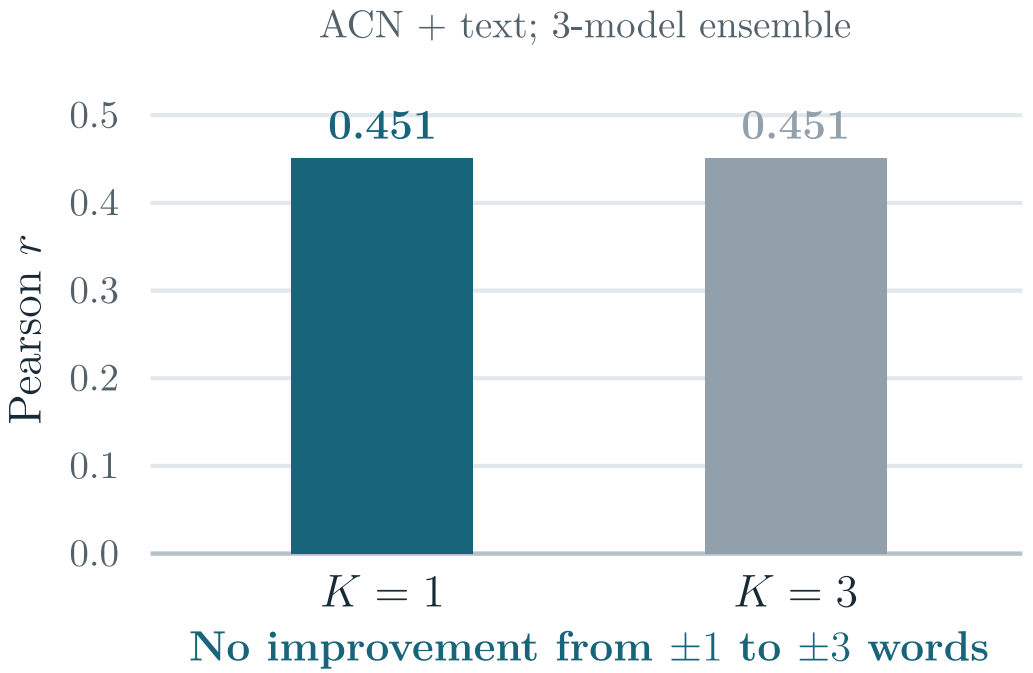
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Preceding-word dominance



96–98% weight on the preceding word

Context-window ablation



No improvement from ±1 to ±3 words

Directional weights are learned constants, independent of the input. The $K = 1$ model still uses the following word.

Context result applies to this cross-corpus evaluation; both values are reported to three decimals.

Limitation

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- Comparisons with distilled or pruned speech Transformers, such as DistilHuBERT and LightHuBERT, remain future work.The learned directional asymmetry and cue weighting require **validation through controlled perceptual experiments**.
- Evaluation is limited to **read** English speech with externally supplied word boundaries, and reliance on following-word features leaves end-to-end streaming latency untested.

Conclusion

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Prominence may be simpler than our models suggest

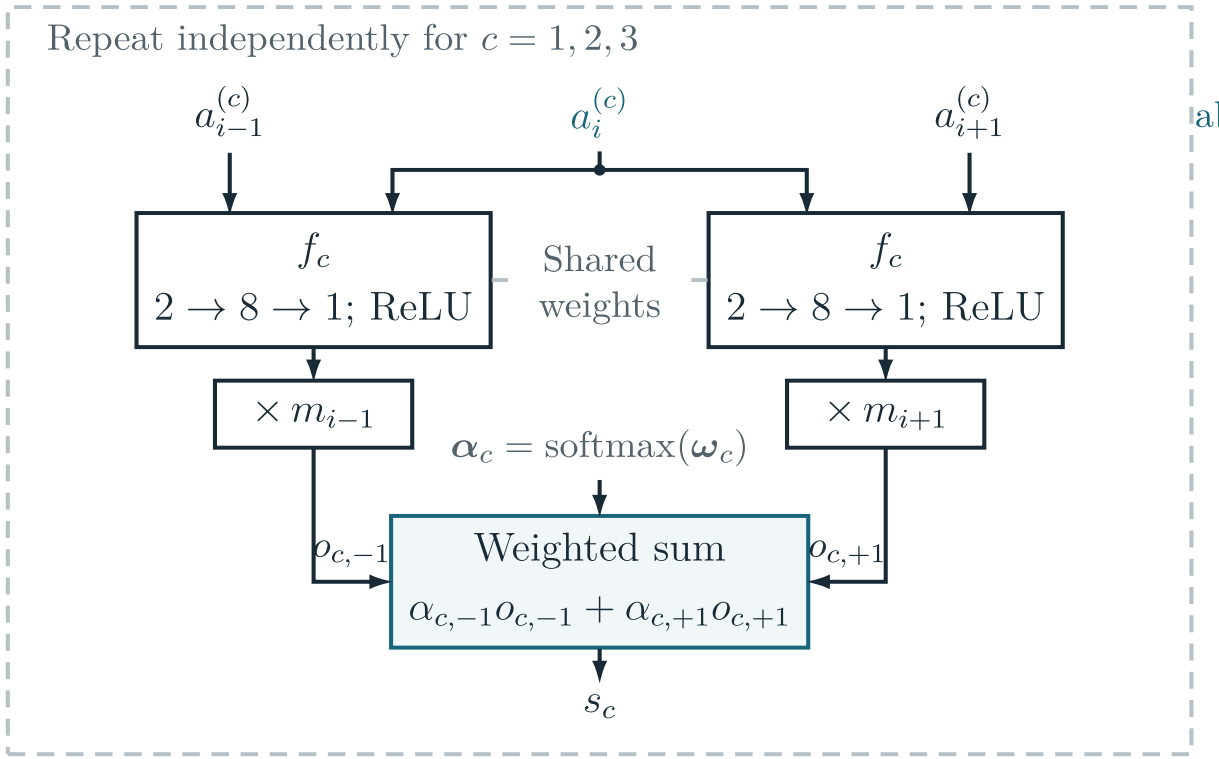
- **Efficient**
  - 202 parameters, 120 $\times$ lower latency
- **Interpretable**
  - Direction, locality, and cue hierarchy are directly observable
- **Text-free(but need)**
  - Suitable when ASR or lexical information is unavailable

Is forward dominance perceptual — or merely statistical?

Appendix

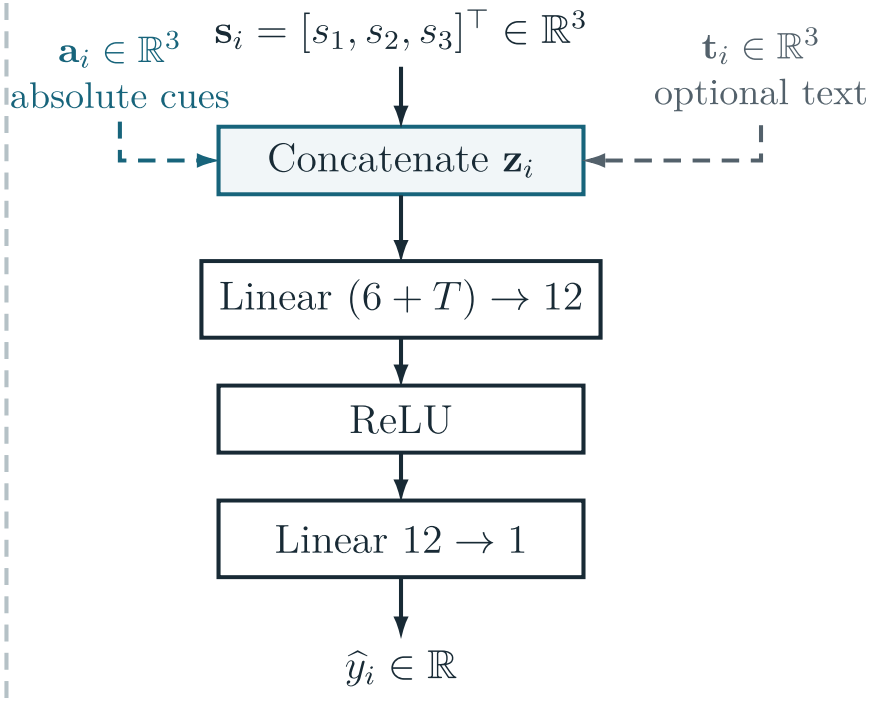
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(a) One cue branch c



Separate MLPs across cues.
Direction weights stay fixed after training.

(b) Aggregation



$T = 0$: acoustic only, 202 parameters.
$T = 3$: with text, 238 parameters.